disease in humans is unknown, but they may account for as many as 1 in 500 mutations. In addition, aberrant recombination events between different LINE repeats or *Alu* repeats can also be a cause of mutation in some genetic diseases (see Chapter 12).

An important additional type of repetitive DNA found in many different locations around the genome includes sequences that are duplicated, often with extraordinarily high sequence conservation. Duplications involving substantial segments of a chromosome, called **segmental duplications,** can span hundreds of kilobase pairs and account for at least 5% of the genome. When the duplicated regions contain genes, genomic rearrangements involving the duplicated sequences can result in the deletion of the region (and the genes) between the copies and thus give rise to disease (see Chapters 5 and 6).

VARIATION IN THE HUMAN GENOME

With completion of the reference human genome sequence, much attention has turned to the discovery and cataloguing of variation in sequence among different individuals (including both healthy individuals and those with various diseases) and among different populations around the globe. As we will explore in much more detail in Chapter 4, there are many tens of millions of common sequence variants that are seen at significant frequency in one or more populations; any given individual carries at least 5 million of these sequence variants. In addition, there are countless very rare variants, many of which probably exist in only a single or a few individuals. In fact, given the number of individuals in our species, *essentially each and every base pair in the human genome is expected to vary in someone somewhere around the globe*. It is for this reason that the original human genome sequence is considered a "reference" sequence for our species, but one that is actually identical to no individual's genome.

Early estimates were that any two randomly selected individuals would have sequences that are 99.9% identical or, put another way, that an individual genome would carry two *different* versions (**alleles**) of the human genome sequence at some 3 to 5 million positions, with different bases (e.g., a T or a G) at the maternally and paternally inherited copies of that particular sequence position (see Fig. 2-6). Although many of these allelic differences involve simply one nucleotide, much of the variation consists of insertions or deletions of (usually) short sequence stretches, variation in the number of copies of repeated elements (including genes), or inversions in the order of sequences at a particular position (**locus**) in the genome (see Chapter 4).

The total amount of the genome involved in such variation is now known to be substantially more than originally estimated and approaches 0.5% between any two randomly selected individuals. As will be addressed in future chapters, any and all of these types of variation can influence biological function and thus must be accounted for in any attempt to understand the contribution of genetics to human health.

TRANSMISSION OF THE GENOME

The chromosomal basis of heredity lies in the copying of the genome and its transmission from a cell to its progeny during typical cell division and from one generation to the next during reproduction, when single copies of the genome from each parent come together in a new embryo.

To achieve these related but distinct forms of genome inheritance, there are two kinds of cell division, mitosis and meiosis. **Mitosis** is ordinary somatic cell division by which the body grows, differentiates, and effects tissue regeneration. Mitotic division normally results in two daughter cells, each with chromosomes and genes identical to those of the parent cell. There may be dozens or even hundreds of successive mitoses in a lineage of somatic cells. In contrast, **meiosis** occurs only in cells of the germline. Meiosis results in the formation of reproductive cells (**gametes**), each of which has only 23 chromosomes—one of each kind of autosome and either an X or a Y. Thus, whereas somatic cells have the **diploid** (*diploos*, double) or the 2n chromosome complement (i.e., 46 chromosomes), gametes have the **haploid** (*haploos*, single) or the n complement (i.e., 23 chromosomes). Abnormalities of chromosome number or structure, which are usually clinically significant, can arise either in somatic cells or in cells of the germline by errors in cell division.

The Cell Cycle

A human being begins life as a fertilized ovum (**zygote**), a diploid cell from which all the cells of the body (estimated to be approximately 100 trillion in number) are derived by a series of dozens or even hundreds of mitoses. Mitosis is obviously crucial for growth and differentiation, but it takes up only a small part of the life cycle of a cell. The period between two successive mitoses is called **interphase,** the state in which most of the life of a cell is spent.

Immediately after mitosis, the cell enters a phase, called G_1, in which there is no DNA synthesis (Fig. 2-8). Some cells pass through this stage in hours; others spend a long time, days or years, in G_1. In fact, some cell types, such as neurons and red blood cells, do not divide at all once they are fully differentiated; rather, they are permanently arrested in a distinct phase known as G_0 ("G zero"). Other cells, such as liver cells, may enter G_0 but, after organ damage, return to G_1 and continue through the cell cycle.

The cell cycle is governed by a series of **checkpoints** that determine the timing of each step in mitosis. In